



POP!WORK: A WEB-BASED GIG PLATFORM

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INTRODUCTION

POPWork (job portal) is the go-to web-based for individuals seeking flexible, short-term job opportunities and for companies needing reliable one-time workers. With this app, users can easily discover, RSVP, and check into temporary job listings across various industries, tailored to fit their availability. Each user gets three free RSVPs, after which they can subscribe to continue accessing more job opportunities. Companies benefit from easy-to-use tools to post jobs, track workers' time with a GPS-based check-in/check-out system and ensure productivity on-site. Perfect for those who want to work on their terms and for companies looking for efficient, on demand staffing.

PROBLEM STATEMENT & OBJECTIVES

Despite the global surge in the gig economy, many existing platforms prioritize long-term employment, creating a critical gap for students and freelancers seeking flexible, short-term work (Smith, 2023). In Sabah, this issue is compounded by the lack of a centralized platform, forcing users to rely on scattered social media listings that are often outdated or fraudulent. Furthermore, the absence of specific flexible work provisions in local labor laws (RSIS International, 2023) and persistent concerns over transparency and income volatility (Ravenelle, 2019) underscore the urgent need for a secure, verified, and student-centric gig ecosystem.

Objective 1: To design a web-based gig platform that connects individuals, particularly students, with short-term job opportunities that align with their availability and lifestyle.

Objective 2: To develop a web-based gig platform using the Agile methodology, incorporating technologies such as PHP, HTML, CSS, JavaScript (Leaflet.js for Maps, FontAwesome for UI), MySQL, XAMPP, and OpenStreetMap API.

Objective 3: To Evaluate the functionality and usability of the system using the System Usability Scale (SUS).

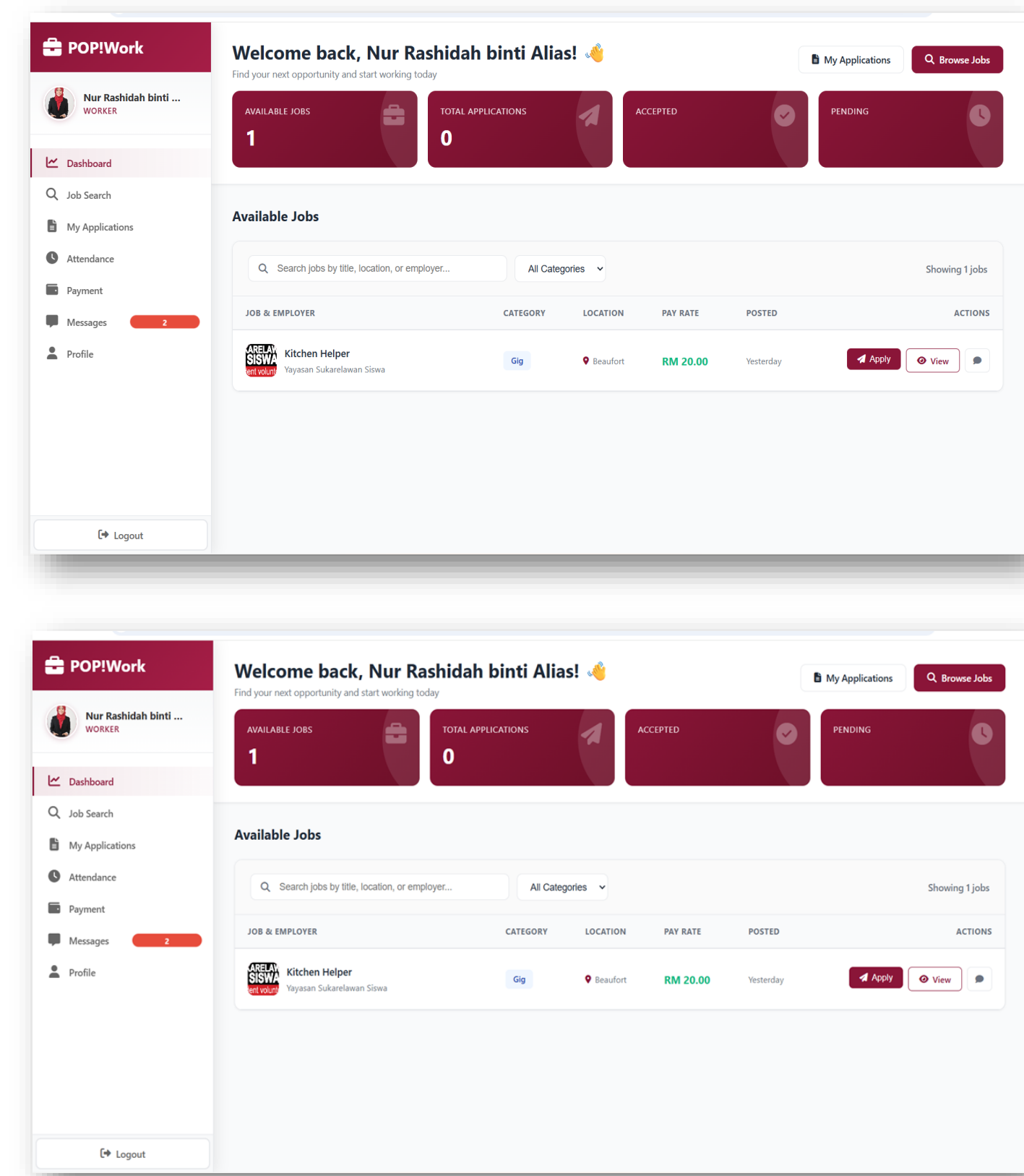


Figure 1: Main System Interface (Jobseeker & Job Offerer)

METHODOLOGY

The POP!WORK platform is built using a robust **Agile Development** lifecycle. The frontend is developed with **React Native** for cross-platform mobile accessibility, enhanced by **Magic Pattern** for distinct UI aesthetics and **Leaflet.js** for geolocation services. The backend relies on **PHP** and an **Apache** server to manage logic, while a normalized **MariaDB** database ensures secure data handling. The system's architecture is defined by a comprehensive **Entity Relationship Diagram (ERD)**

that maps the complex interactions between users, job postings, and payment records, ensuring a seamless experience for the gig economy.

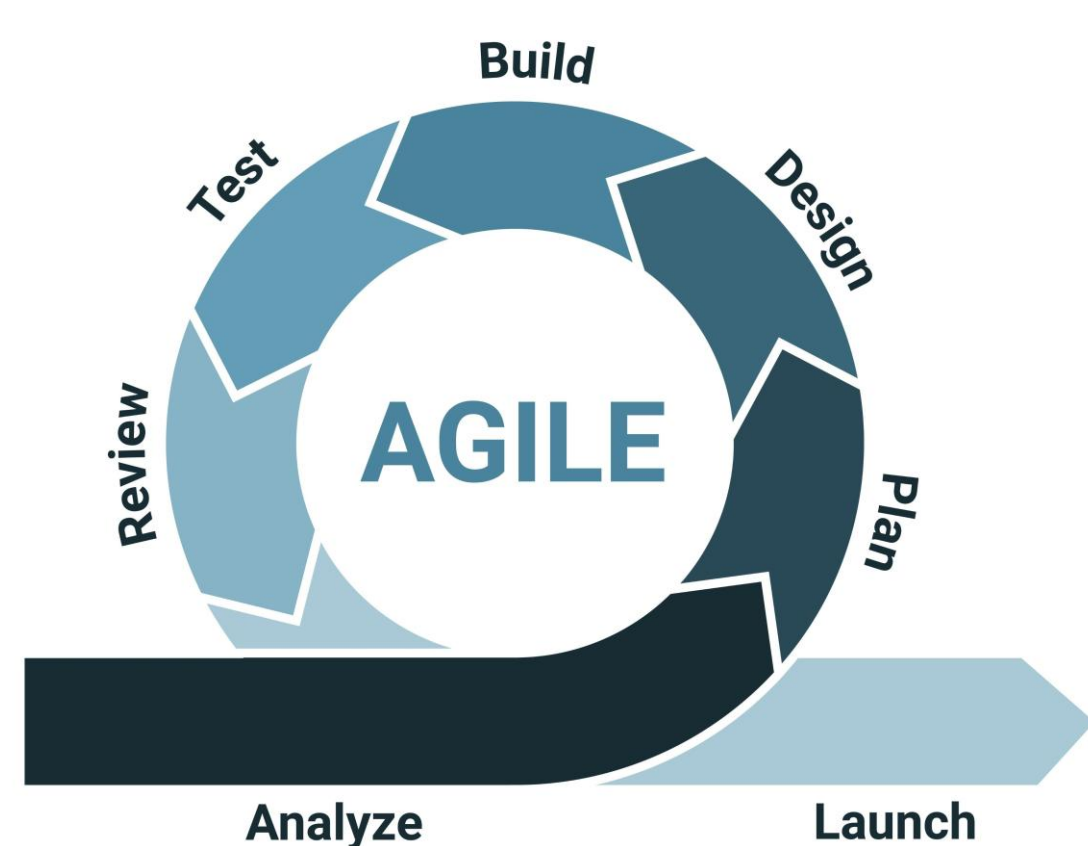
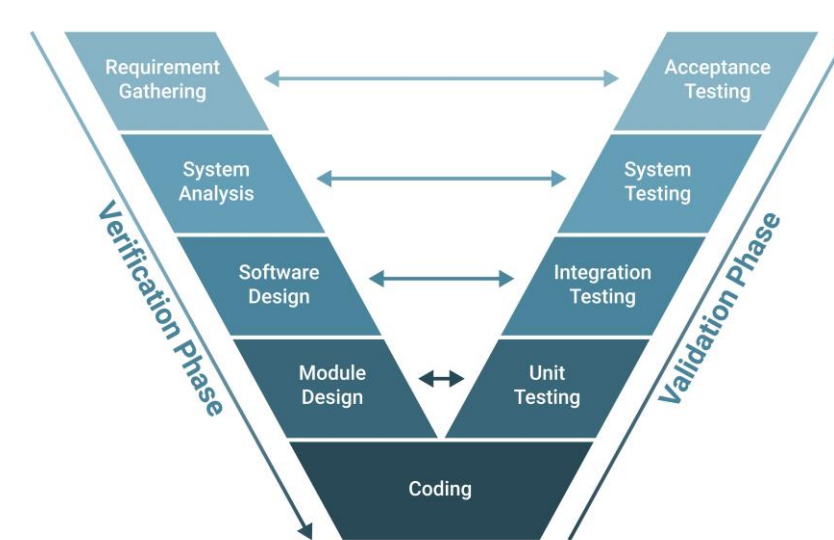


Table 1: Comprehensive Comparative Analysis of Existing Platforms

Feature Category	Feature Description	POPIWORK	Troopers	GoGetter	DoWork	Maukerja	Indeed
1. Platform Architecture	Primary Access	Web-Based	Mobile App	Mobile App	Mobile App	Web-Based	Web-Based
	Installation Required	No (Lightweight)	Yes (Heavy)	Yes	Yes	No	No
2. Target Demographics	Primary Audience	Students (Sabah)	Freelancers	General Public	F&B/Retail	Job Seekers	General Public
	Geographic Focus	Sabah (Local)	Peninsular	Klang Valley	Peninsular	National	Global
3. Job Management	Job Type	Short-term / Gigs	Flexible Gigs	Task / Dispatch	Shifts / Gigs	Full-Time	All Types
	GPS Clock-in/Out	Yes (Integrated)	Yes	Yes	No	No	No
	Real-Time Tracking	Yes	Yes	Yes	No	No	No
4. Safety & Trust	Employer Verification	High (Manual)	High	High	Moderate	Moderate	Low (Aggregator)
	Student Verification	Yes (ID/Matrix)	Yes (IC)	Yes (IC)	Yes	No	No
	Scam Protection	High	High	High	Moderate	Low	Low
5. User Experience	Schedule Alignment	Academic Focused	Shift Based	Task Based	Shift Based	N/A	N/A
	Resume Builder	Auto-Generated	In-App Profile	In-App Profile	Basic Profile	Upload PDF	Upload PDF

SYSTEM VALIDATION



The POP!WORK application was validated through a comprehensive testing framework encompassing Unit, Integration, System, and User Acceptance Testing (UAT).

Integration Testing focused on the interfaces between components, ensuring seamless data connectivity between the MariaDB database, the frontend user interface, and external services like the OpenStreetMap API. **System Testing** verified the application's overall compliance with project requirements, rigorously checking security protocols, session handling, and the accuracy of the GPS geolocation features. Finally, **User Acceptance Testing (UAT)** was performed using the **System Usability Scale (SUS)**, allowing the target demographic to evaluate the platform's usability and effectiveness in a real-world scenario.

MAIN REFERENCE

- Smith, J. (2023). Bridging the gap: Temporary and gig work's role in the modern job market. Journal of Employment Studies, 15(4), 101–115.
- Graham, M., Hjorth, I., & Lehdonvirta, V. (2017). Digital labour and development: impacts of global digital labour platforms and the gig economy on worker livelihoods. Transfer: European Review of Labour and Research, 23(2), 135–162. <https://doi.org/10.1177/1024258916687250>

SYSTEM SCOPE & DESIGN

User Registration and Profile Management allows students and employers to create secure accounts and set up comprehensive profiles. It allows students to upload resumes, list skills, and manage their availability, while allowing employers to set up business profiles and verify their identity. Users can also modify their profile information and deactivate their accounts if necessary.

Job Posting and Management allows employers to create detailed job vacancies including title, description, pay rate, date, and specific location coordinates. It allows employers to edit job details, close vacancies once filled, and remove outdated listings. It also provides a dashboard for employers to review incoming applications and approve or reject candidates.

Job Discovery and Application allows student workers to browse available gigs and specify search criteria based on location, category, and pay rate. It allows users to view detailed job descriptions and locations via an interactive map before applying. Users can submit applications for jobs and track the status of their applications in real-time.

GPS Attendance and Verification allows workers to perform "Clock-In" and "Clock-Out" actions only when they are within the specific GPS radius of the job site. This module validates the user's physical location to ensure attendance integrity and allows employers to monitor live attendance status to prevent absenteeism.

Payment and History Records allows employers to record payment transactions upon job completion and rate the worker's performance. It allows students to view their work history, track total earnings, and receive ratings, creating a verified digital resume of their gig work experience.

Admin and Content Moderation allows administrators to oversee the entire ecosystem. It allows admins to verify new employer accounts to prevent fraud, remove inappropriate or suspicious job postings, and ban users who violate platform policies. This ensures the platform remains a safe environment for the Sabah student community.

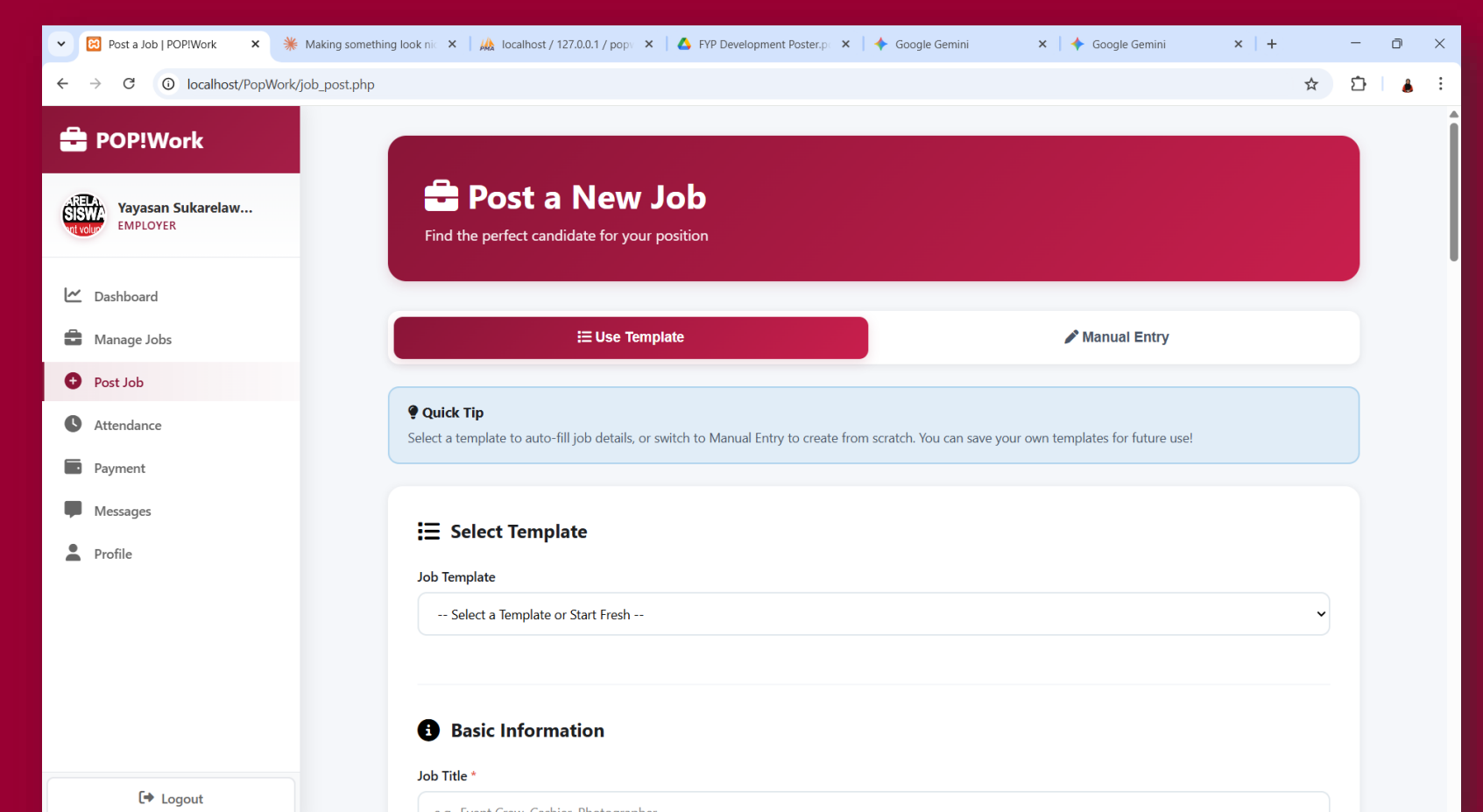


Figure 2: Post a Job Interface

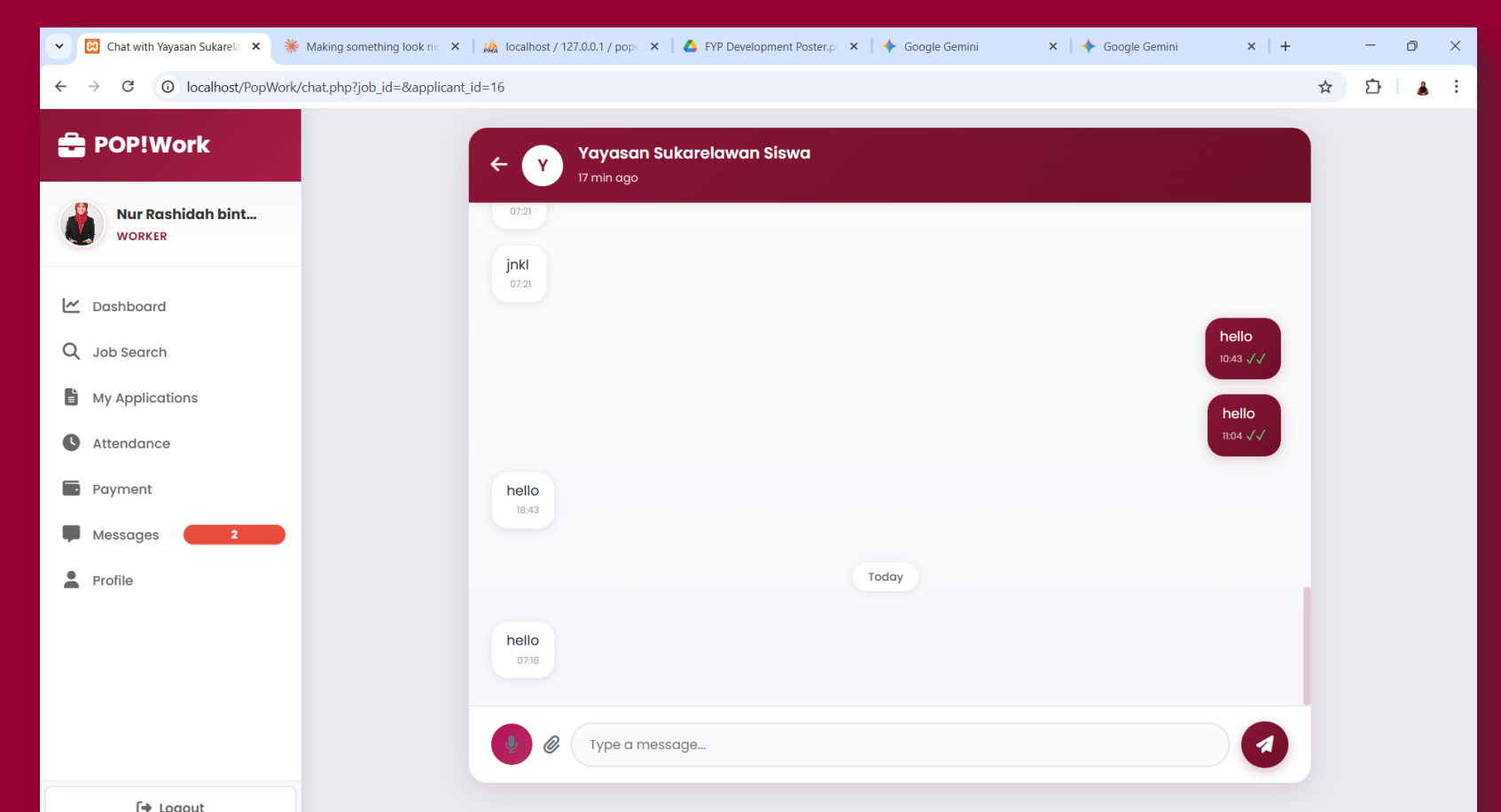


Figure 3: Chat Interface